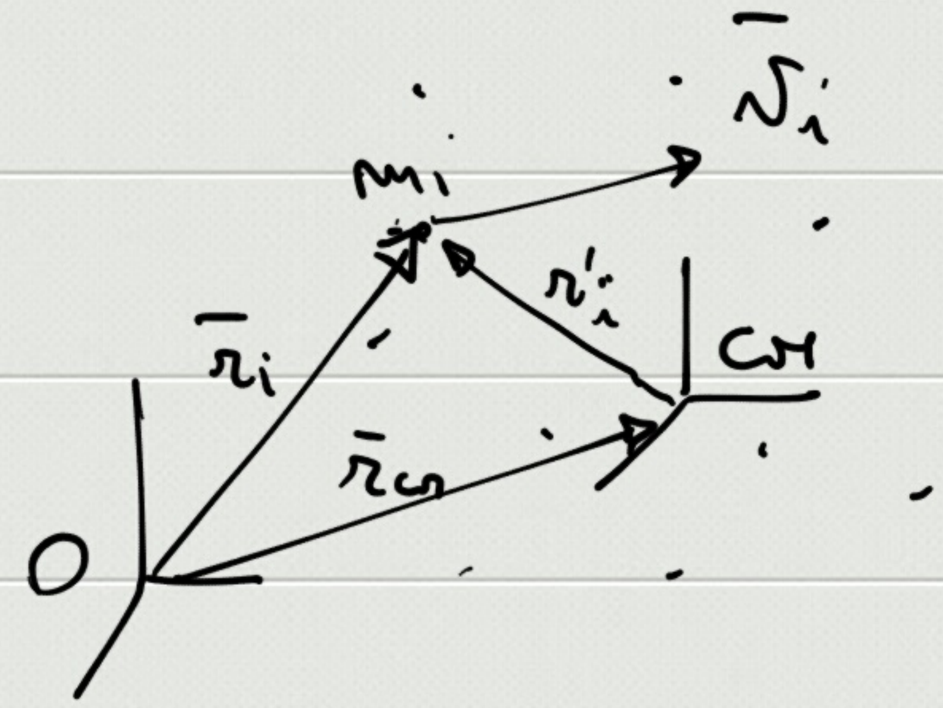
$$\bar{L}_O = \sum_i \bar{L}_{O,i} = \sum_i \bar{r}_i \times m_i \bar{v}_i$$



$$\bar{r}_i = \bar{r}'_i + \bar{r}_{CM} \qquad \bar{v}_i = \bar{v}'_i + \bar{v}_{CM}$$

$$\Rightarrow \bar{L}_O = \sum_i (\bar{r}'_i + \bar{r}_{CM}) \times m_i (\bar{v}'_i + \bar{v}_{CM}) =$$

$$= \sum_i \bar{r}'_i \times m_i \bar{v}'_i + \sum_i \bar{r}'_i \times m_i \bar{v}_{CM} +$$

$$+ \sum_i \bar{r}_{CM} \times m_i \bar{v}'_i + \sum_i \bar{r}_{CM} \times m_i \bar{v}_{CM} =$$

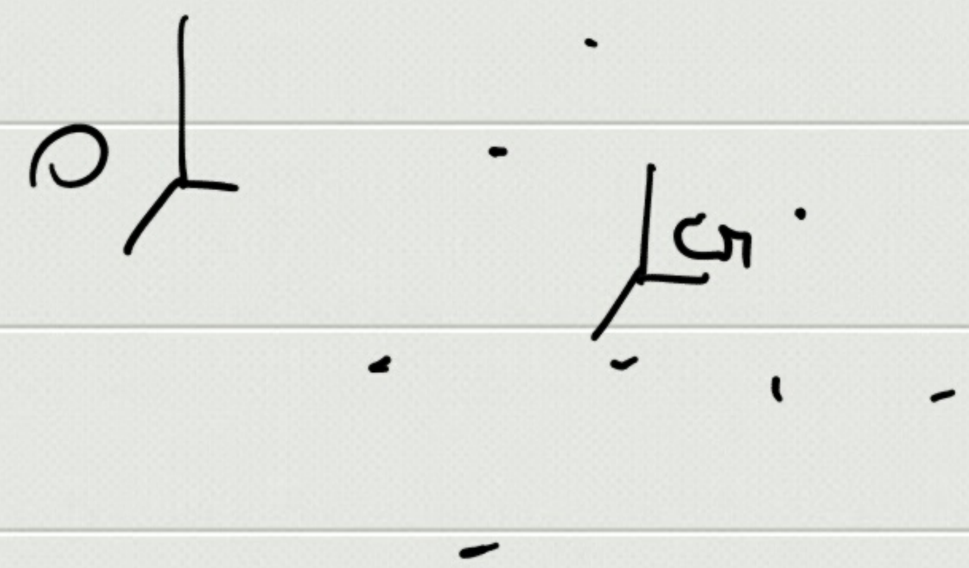
$$= \sum_i \bar{L}'_{CM,i} + \cancel{\Big(\underbrace{\sum_i m_i \bar{r}'_i}_{=0}\Big) \times \bar{v}_{CM}} + \bar{r}_{CM} \times \cancel{\Big(\underbrace{\sum_i m_i \bar{v}'_i}_{=0}\Big)} + \bar{r}_{CM} \times \sum_i m_i \bar{v}_{CM} =$$

$$= \bar{L}'_{CM} + \bar{r}_{CM} \times m_{TOT} \bar{v}_{CM}$$

$$\boxed{\bar{L}_O = \bar{L}'_{CM} + \bar{L}_{O,CM}}$$

T. König per il momento angolare

$\bar{v}_i = \bar{v}_i' + \bar{v}_{CM}$



$$E_k = \sum_i E_{k,i} = \sum_i \frac{1}{2} m_i v_i^2 = \qquad v_i^2 = \bar{v}_i \cdot \bar{v}_i$$

$$= \sum_i \frac{1}{2} m_i \left(\bar{v}_i' + \bar{v}_{CM}\right)^2 =$$

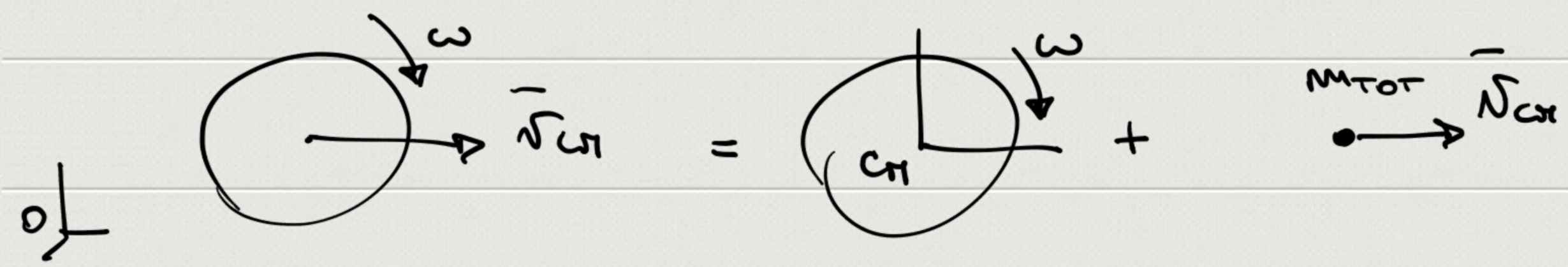
$$= \sum_i \frac{1}{2} m_i v_i'^2 + \sum_i \frac{1}{2} m_i v_{CM}^2 + \sum_i \frac{\cancel{2}}{\cancel{2}} m_i \cdot \cancel{2}\, \bar{v}_i' \cdot \bar{v}_{CM} =$$

$$= \sum_i E_{k,i}' + \frac{1}{2} m_{TOT} v_{CM}^2 + \underbrace{\left(\sum_i m_i \bar{v}_i'\right)}_{=0} \cdot \bar{v}_{CM}$$

$$= E_k' + \frac{1}{2} m_{TOT} v_{CM}^2$$

$$\boxed{E_k = E_k' + E_{k,CM}}$$

T. Königs per l'energia cinetica



$$E_k = E_k' + E_{k,CM}$$

SdR del CM

$$\bar{P} = m_{TOT}\,\bar{v}_{CM} \qquad (\bar{P}' = 0)$$

$$\bar{R}^{E} = m_{TOT}\,\bar{a}_{CM} \qquad (\bar{R}^{E} = 0)$$

moto di traslazione

$$\bar{L}_0 = \bar{L}'_{CM} + \bar{L}_{0,CM}$$

$$\bar{M}^{E}_0 = \bar{M}'^{E}_{CM} + \bar{M}_{0,CM}$$

$$E_K = E'_K + E_{K,CM}$$

$\Rightarrow$ moto di rotazione